

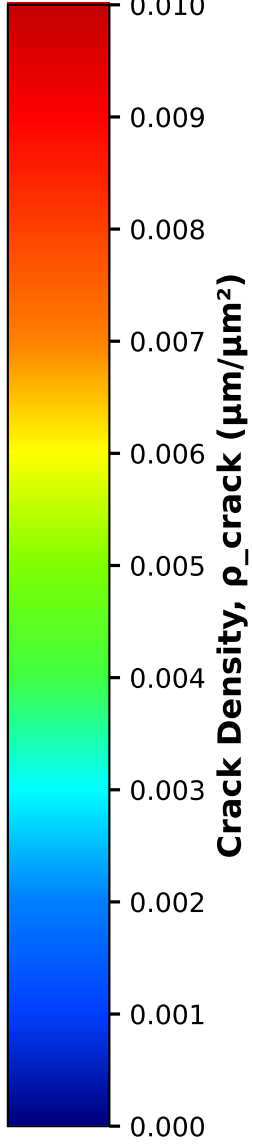
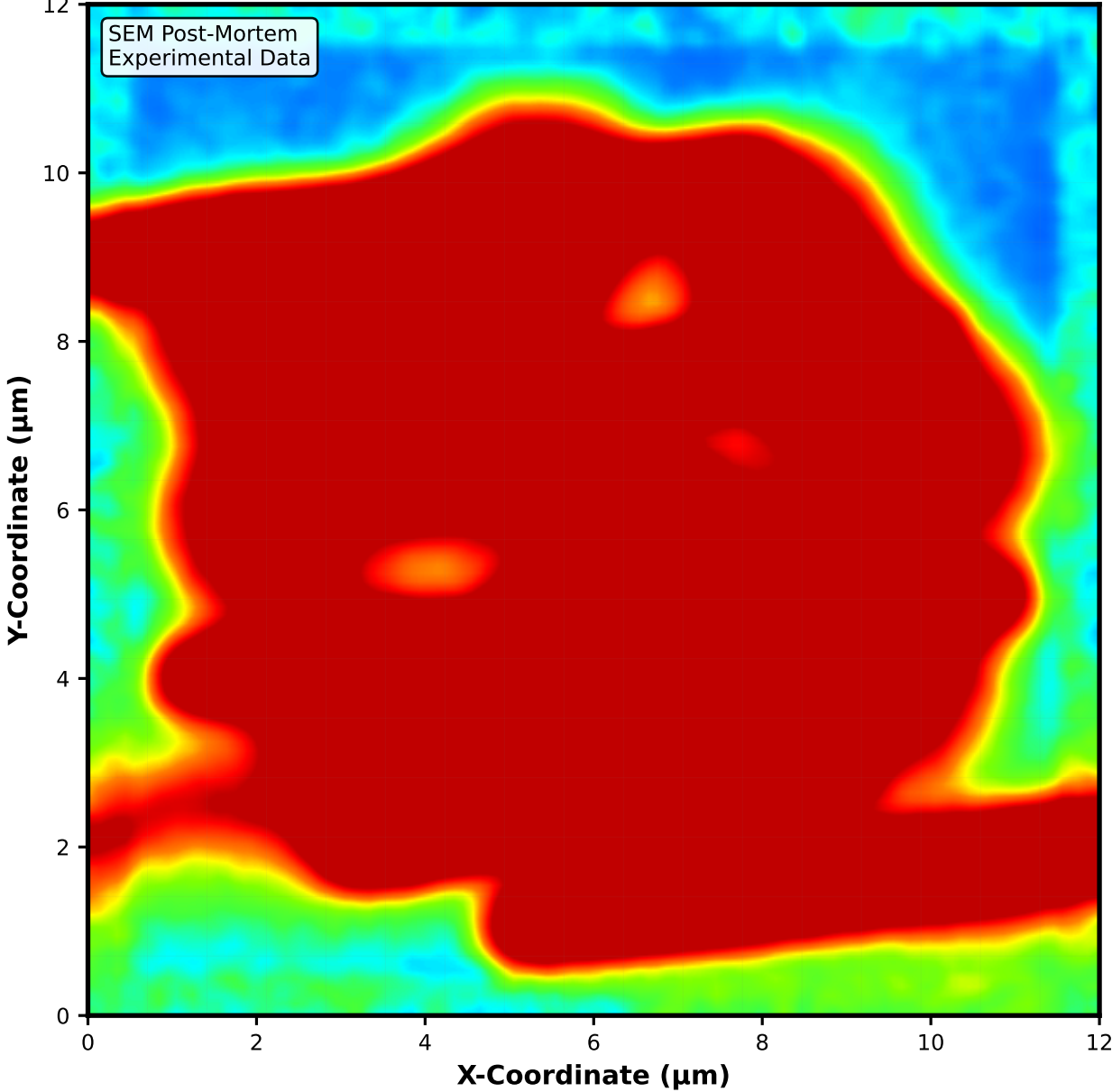
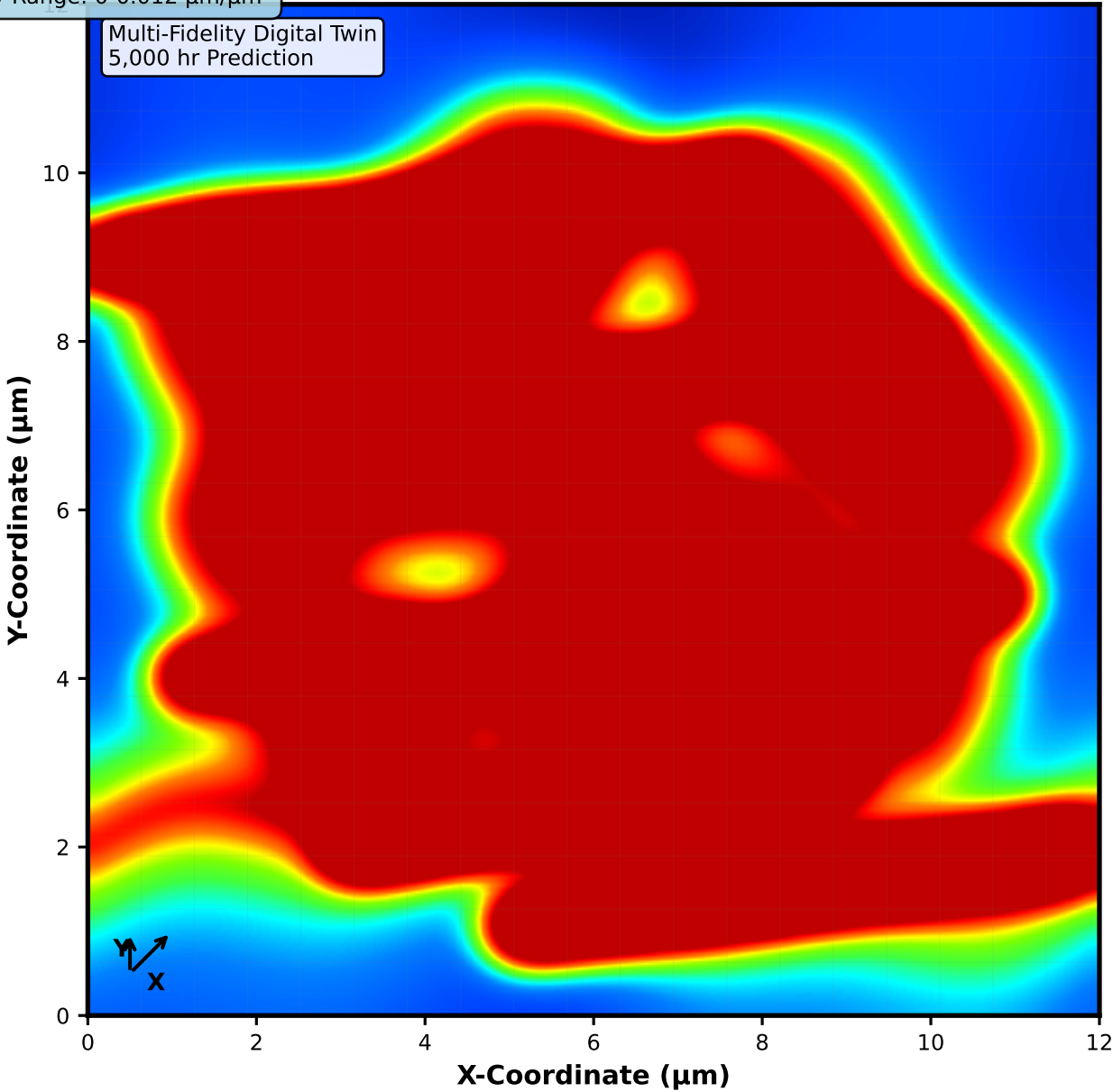
- Analysis Specifications:
- Domain: 12 × 12 μm SOFC anode
 - Operating Time: 5,000 hours
 - Temperature: 800°C
 - Spatial Resolution: 0.1 μm
 - Crack Density Range: 0-0.012 μm/μm²

(a) MF-DL Prediction

Spatial Accuracy of Long-Term Degradation Prognosis:
MF-DL Prediction vs. Experimental Validation

(b) SEM Experimental Data

- Validation Results:
- Spatial Correlation: 0.996
 - Hotspot Accuracy: 94.1%
 - Max Crack Density: 0.0120 μm/μm²
 - Hotspot Threshold: 0.005 μm/μm²



Spatial Correlation Coefficient: 0.996